Week-1
Mathematics for Data Science - 2
Vectors and Matrices
Practice Assignment

1. Match the systems of linear equations in Column A with their number of solutions in Column B and their geometric representation in Column C.

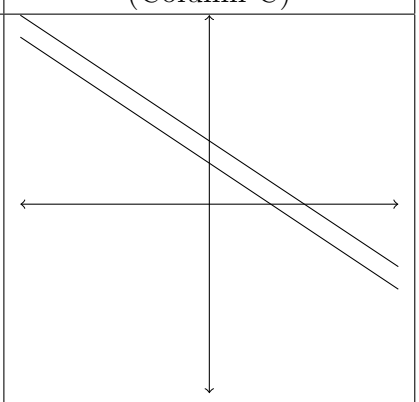
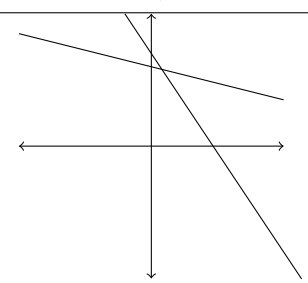
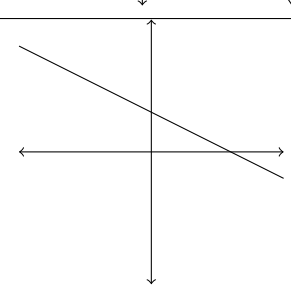
	System of linear equations (Column A)		Number of solutions (Column B)		Geometric representations (Column C)
i)	$3x + 2y = 7,\ x + 4y = 12$	a)	Infinite solutions	1)	
ii)	$2x + 3y = 5,\ 8x + 12y = 13$	b)	No solution	2)	
iii)	$x + 2y = 3,\ 4x + 8y = 12$	c)	Unique solution	3)	

Table: M2W1PT1

- ◯ **Option 1:** i) → c) → 2)
- ◯ Option 2: i) → a) → 1)

- ◯ Option 3: ii) $\rightarrow$ c) $\rightarrow$ 3)
- ◯ Option 4: iii) $\rightarrow$ a) $\rightarrow$ 2)
- ◯ **Option 5:** ii) $\rightarrow$ b) $\rightarrow$ 1)

Solution:

i) Given

$$3x + 2y = 7 \tag{1}$$

$$x + 4y = 12 \tag{2}$$

a system of linear equations in two unknowns (variables) x and y.

Solving these equations we get $x = \frac{2}{5}$, $y = \frac{29}{10}$. (i.e., multiplying (2) by 3 and subtracting the resultant equation from (1), we get $y = \frac{29}{10}$. Substituting $y = \frac{29}{10}$ in the equation (2) we have $x = \frac{2}{5}$.)

As we get only one pair of values for x and y i.e., these two straight lines are intersecting at a unique point so the given system of linear equations has a unique solution.
So from Column (B) the option (c) is correct.

Now, in $\mathbb{R}^2$ both the equations represent straight lines and they intersect each other at $x = \frac{2}{5}$, $y = \frac{29}{10}$ i.e., $(\frac{2}{5}, \frac{29}{10})$. So from Column (C) the option (2) is correct.

Hence option 1: i) $\rightarrow$ c) $\rightarrow$ 2) is correct.

ii) Given

$$2x + 3y = 5 \tag{3}$$

$$8x + 3y = 13 \tag{4}$$

a system of linear equations in two unknowns (variables) x and y.

Multiplying (3) by 4 and subtracting the resultant equation from (4), we get $0 = -7$, which is absurd. Hence no solution exists. i.e., the straight lines are parallel.
So from Column (B) the option (b) is correct, from Column (C) the option (1) is correct.

Hence option 5: ii) $\rightarrow$ b) $\rightarrow$ 1) is correct.

iii) Given

$$x + 2y = 3 \tag{5}$$

$$4x + 8y = 12 \tag{6}$$

a system of linear equations in two unknowns (variables) x and y. Multiplying (5) by 4, we get the equation (6) So, the equations (5) and (6) are basically same so there are infinite number of solutions. These two equations represent the same straight line. Hence no solution exists. i.e., the straight lines are parallel.
So from Column (B), option (a) is correct but from Column (C) no option represents the same line so no option is correct.

Hence from the given options no option is correct.

2. Choose the set of correct options

- ◯ **Option 1:** If both A and B are 2×2 real matrices and $det(AB) = 0$, then $det(A) = 0$ or $det(B) = 0$.
- ◯ **Option 2:** If A is a 3×3 real matrix with non-zero determinant and k is some real number, then $det(k\ A) = k^3 \times det(A)$.
- ◯ **Option 3:** If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$, then $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$ (where A^n is the matrix $A \times A \times \ldots \times A$, n-times).
- ◯ **Option 4:** The number of scalar additions to be done to compute the matrix AB, where A is a 3×2 matrix and B is a 2×3 matrix, is 9.

Solution:
Option 1: From properties of determinant we have "for any two square matrices of order n, the determinant of the product of two matrices is equal to the product of their determinants" i.e., $det(AB) = det(A)\ det(B)$

Hence $det(AB) = 0 \implies det(A)det(B) = 0 \implies det(A) = 0$ or $det(B) = 0$.

Hence option 1 is correct.

Option 2: By the property of determinant "If A is a square matrix of order $n \times n$, then $det(cA) = c^n\ det(A)$."

OR

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

then $det(A) = a(ei - hf) - b(di - gf) + c(dh - ge)$

Now, $kA = \begin{bmatrix} ka & kb & kc \\ kd & ke & kf \\ kg & kh & ki \end{bmatrix}$

so, $det(kA) = ka(keki - kh.kf) - kb(kdki - kgkf) + kc(kdkh - kgke)$

$$\begin{aligned} det(kA) =& ka(keki - kh.kf) - kb(kdki - kgkf) + kc(kdkh - kgke) \\ =& k^3[a(ei - hf) - b(di - gf) + c(dh - ge)] \\ =& k^3 det(A) \end{aligned}$$

Hence option 2 is correct.

Option 3: Given $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Now, $A^2 = 2 \times 2 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 2^2 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Now, $A^3 = A^2 \times A = 2^2 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \times 2 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 2^3 \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ and so on.

By induction we have $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$

Hence option 3 is correct.

Option 4: Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$ *and* $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$ be two matrices of order 3×2 and 2×3 respectively.

Now $AB = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$

$= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} & a_{31}b_{13} + a_{32}b_{23} \end{bmatrix}$

Hence there are 9 scalar additions.

3. Choose the set of correct options

   - ◯ **Option 1:** A triangular 3×3 matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
   - ◯ Option 2: If A and B are 3×3 matrices then $det(A + B) = det(A) + det(B)$.
   - ◯ Option 3: If A and B are 3×3 matrices then $adj(A + B) = adj(A) + adj(B)$.
   - ◯ **Option 4:** If A is a 3×3 matrix and B is a matrix obtained from A by multiplying each column of A by its column number, then $det(B) = 6det(A)$.
   - ◯ **Option 5:** If the sum of the first and the third row vectors of a 3×3 matrix A is equal to the second row vector of A, then $det(A) = 0$.

Solution:
Option 1: The determinant of a triangular matrix is product of diagonal elements. Hence the determinant is non-zero if and only if all diagonal elements are non-zero.

Hence option 1 is correct.

Option 2: Let $A = B$ be the identity I matrix of order 3 i.e., $A = I_{3\times3} = B$.

So, $A + B = I_{3\times3} + I_{3\times3} = 2I_{3\times3}$

Now $det(A + B) = det(2I_{3\times3}) = 2^3 det(I_{3\times3}) = 8 \times 1 = 8$

$det(A) + det(B) = det(I_{3\times3}) + det(I_{3\times3}) = 1 + 1 = 2$

Hence $det(A + B) \neq det(A) + det(B)$.

Hence option 2 is not correct.

Option 3: Let $A = I_{3\times3} = B$.

So, $A + B = I_{3\times3} + I_{3\times3} = 2I_{3\times3}$.

Now $adj(A) = I_{3\times3}$, $adj(B) = I_{3\times3}$, $adj(A) + adj(B) = I_{3\times3} + I_{3\times3} = 2I_{3\times3}$ and $adj(A + B) = 4I_{3\times3}$

So, $adj(A + B) \neq adj(A) + adj(B)$.

Hence option 3 is not correct.

Option 4: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} a_{11} & 2a_{12} & 3a_{13} \\ a_{21} & 2a_{22} & 3a_{23} \\ a_{31} & 2a_{32} & 3a_{33} \end{bmatrix}$

$det(A) = a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$

Now $det(B) = a_{11}(2a_{22}3a_{33} - 2a_{32}3a_{23}) - 2a_{12}(a_{21}3a_{33} - a_{31}3a_{23}) + a_{13}(a_{21}2a_{32} - a_{31}2a_{22})$

$det(B) = 6[a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$

$det(B) = 6\ det(A)$

Hence option 4 is correct.

Option 5: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} + a_{31} & a_{12} + a_{32} & a_{13} + a_{33} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Applying the row transformation $R_2 \to (R_1 + R_3)$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$det(A) = 0$.

Hence option 5 is correct.

4. Mahesh bought 2 kg potato and c kg dal from a shop, and paid ₹200 to the shopkeeper. Gaurav bought 4 kg potato and 4 kg dal, and paid ₹d to the shopkeeper. If x_1 represents the price of 1 kg potato and x_2 represents the price of 1 kg dal, then choose the set of correct options.

   ◯ Option 1: The matrix representation to find x_1 and x_2 is

$$\begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$$

- ◯ **Option 2:** The matrix representation to find x_1 and x_2 is

$$\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$$

- ◯ **Option 3:** If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will not be unique.
- ◯ Option 4: If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will be unique.
- ◯ Option 5: If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d \neq 400$, then he will be able to find the price (as a numerical value) of 1 kg potato.
- ◯ **Option 6:** If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d \neq 400$, then he will fail to find the price (as a numerical value) of 1 kg potato.

Solution: Let x_1 represent the price of 1 kg potato and x_2 represent the price of 1 kg dal. Given that Mahesh bought 2 kg potato and c kg dal from a shop, and paid ₹200 to the shopkeeper. So from this given information we have $2x_1 + cx_2 = 200$.

Also, Gaurav bought 4 kg potato and 4 kg dal, and paid ₹d to the shopkeeper. So, $4x_1 + 4x_2 = d$.

These form a system of linear equations and its matrix representation is

$$\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$$

Hence option 2 is correct.

Let Mahesh try to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$. Then the corresponding equations are $2x_1 + 2x_2 = 200$, $4x_1 + 4x_2 = 400$.

We observed that the second equation is a scalar multiple of the first equation i.e., they represent the same equation $x_1 + x_2 = 100$ so it has infinite number of solutions. Hence the solution is not unique.

Hence option 3 is correct.

Let Mahesh try to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d \neq 400$. Then the corresponding equations are $2x_1 + 2x_2 = 200$, $4x_1 + 4x_2 = d$.

So we have $x_1 + x_2 = 100$, $x_1 + x_2 = \frac{d}{4} \implies 100 = \frac{d}{4} \implies d = 400$, which contradicts our assumptions, so there is no solution.

Hence option 6 is correct.

5. The marks obtained by Safina, Ram and Pratiksha in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table W1PT2.

	Quiz 1	Quiz 2	End sem
Safina	89	95	88
Ram	92	81	98
Pratiksha	85	93	98

Table: W1PT2

The weightage of marks in final grade(in percent) of Quiz 1, Quiz 2, and End sem is shown in Table W1PT3.

	In percent (%)
Quiz 1	20
Quiz 2	20
End sem	60

Table: W1PT3

Choose the set of correct options.

- ◯ Option 1: Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix:
$$\begin{bmatrix} 93.4 \\ 94.4 \\ 89.6 \end{bmatrix}$$
- ◯ **Option 2:** Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :
$$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$$
- ◯ **Option 3:** The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is 3×1

◯ **Option 4:** If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$ then the overall final grades (in 100) can be represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

Solution: The matrix representation of the marks obtained by these students and the weightage are $\begin{bmatrix} 89 & 95 & 88 \\ 92 & 81 & 98 \\ 85 & 93 & 98 \end{bmatrix}$ and $\begin{bmatrix} \frac{20}{100} \\ \frac{20}{100} \\ \frac{60}{100} \end{bmatrix}$

Hence the final grade represented by the vector is $\begin{bmatrix} 89 \times \frac{20}{100} + 95 \times \frac{20}{100} + 88 \times \frac{60}{100} \\ 92 \times \frac{20}{100} + 81 \times \frac{20}{100} + 98 \times \frac{60}{100} \\ 85 \times \frac{20}{100} + 93 \times \frac{20}{100} + 98 \times \frac{60}{100} \end{bmatrix} = \begin{bmatrix} 89.6 \\ 93.4 \\ 94.6 \end{bmatrix}$

Hence option 2 and 3 are correct.

Let the bonus marks given to Safina, Ram and Pratiksha be represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$. Then the overall final grades (in 100) will be

$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.6 \end{bmatrix} + \begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix} = \begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$.

Hence option 4 is correct.

6. Let A be a 3×3 matrix with non-zero determinant and B be a matrix obtained by adding 5 times of first row of A to the third row of A and adding 10 times of second row of A to the first row of A. What is the value of $det(2AB^{-1})$? [Ans: 8]

Solution: Let A be a 3×3 matrix with non-zero determinant and B be a matrix obtained by adding 5 times of first row of A to the third row of A and adding 10 times of second row of A to the first row of A. Here the matrix B is obtained by performing elementary row transformation on A, then $det(B) = det(A)$. Now,

$$det(2AB^{-1}) = 2^3 det(A) det(B^{-1}) = 2^3 det(A) \frac{1}{det(B)} = 8 \frac{det(A)}{det(A)} = 8$$

7. If $A = \begin{bmatrix} 2x & 2z & y+x \\ -x+z & 2x & -y+z \\ x+y-2z & y-z-2x & 2y-2z \end{bmatrix}$, then find $det(A)$.

[Ans: $(x-y)(y-z)(z+x)$]

Solution: $det(A) = \begin{vmatrix} 2x & 2z & y+x \\ -x+z & 2x & -y+z \\ x+y-2z & y-z-2x & 2y-2z \end{vmatrix}$

$\xrightarrow{R_3+R_2} \sim \begin{vmatrix} 2x & 2z & y+x \\ -x+z & 2x & -y+z \\ y-z & y-z & y-z \end{vmatrix} \xrightarrow{R_2+R_1} \sim \begin{vmatrix} 2x & 2z & y+x \\ x+z & 2(x+z) & x+z \\ y-z & y-z & y-z \end{vmatrix}$

$\sim (y-z)(x+y) \begin{vmatrix} 2x & 2z & y+x \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = (y-z)(x+y)(2x-0+(y+x)(-1)) = (y-z)(x+y)(x-y)$

Comprehension Type Question:

A shopkeeper sells three types of clothes- shirts, jeans, and T- shirts- in three different sizes: small, medium, and large. In a week, he sold 1 small, 1 medium and 2 large sized shirts; 2 small, c medium and 6 large sized jeans, and 1 small, 3 medium and $c-5$ large sized T-shirts (where c is an integer). The price of shirts, jeans, and T-shirts remain the same for different sizes (i.e., small, medium, and large sized shirts have the same price; similarly, small, medium, large sized jeans have the same price; and small, medium, large sized T-shirts have same price). The shopkeeper earned ₹7 , ₹27 and ₹43 (in thousand) in that week, for small, medium, and large sized clothes respectively.
Answer the following questions using the given data.

8. Let s, j, t represents the price (in thousand) of 1 shirt, 1 jeans and 1 T-shirt respectively and we want to find s, j, t by solving a system of linear equations represented by the matrix form $Ax = b$, where $x = (s, j, t)^T$ and $b = (7, 27, 43)^T$. If sum of entries in the first row of A is p, sum of entries in the second row of A is q and sum of entries in the third row of A is r (take the topmost row as the fist row), then find the value of $p+q-r$.
[Ans: 5]

Solution: From the given data we have,

	Shirt	Jeans	T-Shirts
Small	1	2	1
Medium	1	c	3
Large	2	6	c-5

Let s, j, t represent the price (in thousand) of 1 shirt, 1 jeans and 1 T-shirt respectively.

Hence the matrix representation is $\begin{bmatrix} 1 & 2 & 1 \\ 1 & c & 3 \\ 2 & 6 & c-5 \end{bmatrix} \begin{bmatrix} s \\ j \\ t \end{bmatrix} = \begin{bmatrix} 7 \\ 27 \\ 43 \end{bmatrix}$.

Given sum of entries in the first row of A is p, sum of entries in the second row of A is q and sum of entries in the third row of A is r then $p = 4$, $q = c + 4$ and $r = c + 3$. Thus $p + q - r = 4 + c + 4 - c - 3 = 5$.

9. If $det(A) = 122$, then how many medium sized jeans were sold in the week? [Ans : 16]

   Solution : Let $det(A) = 122$

   $det(A) = \begin{vmatrix} 1 & 2 & 1 \\ 1 & c & 3 \\ 2 & 6 & c-5 \end{vmatrix} = 1(c^2 - 5c - 18) - 2(c - 5 - 6) + 1(6 - 2c)$

   $det(A) = c^2 - 9c + 10 = 122 \implies c^2 - 9c - 112 = 0 \implies (c + 7)(c - 16) = 0.$

   $c = 16$ as c can not be negative.

   Hence 16 many medium sized jeans were sold in the week.

10. If A is the matrix as above (in question 8) and $det(A) = 122$, then what is the price(in thousand) of a T-shirt, where price (in thousand) of a shirt is 2 and price (in thousand) of a jeans is 1? [Ans: 3]

    Solution:

    From the given information we have, $\begin{bmatrix} 1 & 2 & 1 \\ 1 & c & 3 \\ 2 & 6 & c-5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ t \end{bmatrix} = \begin{bmatrix} 7 \\ 27 \\ 43 \end{bmatrix}$

    $2 + 2 + t = 7 \implies t = 3$ which also satisfies the other two equations

    Hence $t = 3$.